Vector Spaces III: Sums, Intersections and Direct Sums

Problem sheet for the Day 4 class · CSIR NET / GATE / IIT JAM

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2013 NAT Let $W_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = x_2 + 2x_3\}$ and $W_2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 3x_2 + 2x_3\}$. Then $\dim(W_1 + W_2) =$ _____.

2. JAM 2019 NAT Let $W_1 = \{x \in \mathbb{R}^{100} : x_i = 0 \text{ whenever } 4 \mid i\}$ and $W_2 = \{x \in \mathbb{R}^{100} : x_i = 0 \text{ whenever } 5 \mid i\}$. Then $\dim(W_1 \cap W_2) =$ _____.

3. Exam pattern MSQ Let S, T be subspaces of $\mathbb{R}^{24}$ with $\dim S = 19$ and $\dim T = 17$. Which of the following are true?

(A) the smallest possible $\dim(S \cap T)$ is 2
(B) the largest possible $\dim(S \cap T)$ is 18
(C) the smallest possible $\dim(S + T)$ is 19
(D) the largest possible $\dim(S + T)$ is 24

4. Exam pattern MSQ In $M_n(\mathbb{R})$, $n \geq 2$, which pairs of subspaces give $M_n(\mathbb{R}) = W_1 \oplus W_2$?

(A) symmetric, skew-symmetric
(B) upper triangular, lower triangular
(C) upper triangular, strictly lower triangular
(D) trace-zero, scalar matrices

5. Exam pattern NAT Let $W_1 = \text{span}\{(1,0,1,0), (0,1,0,1)\}$ and $W_2 = \text{span}\{(1,1,0,0), (0,0,1,1)\}$ in $\mathbb{R}^4$. Then $\dim(W_1 \cap W_2) =$ _____.

6. Exam pattern MSQ Let $W_1 \neq W_2$ be 3-dimensional subspaces of $\mathbb{R}^5$. Then $\dim(W_1 \cap W_2)$ can be

(A) 0
(B) 1
(C) 2
(D) 3

7. Exam pattern MCQ Let W_1, W_2 be subspaces of V. Then $W_1 \cup W_2$ is a subspace

(A) always
(B) never
(C) if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$
(D) if and only if $W_1 \cap W_2 = \{0\}$

8. Exam pattern MSQ Let $W = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$. Which of the following lines are complements of W in $\mathbb{R}^4$?

(A) $\text{span}\{(1,1,1,1)\}$
(B) $\text{span}\{(1,-1,0,0)\}$
(C) $\text{span}\{(1,0,0,0)\}$
(D) $\text{span}\{(1,1,-1,-1)\}$

9. Exam pattern MCQ Let L_1, L_2, L_3 be three distinct lines through the origin in $\mathbb{R}^2$. Which statement is true?

(A) $L_1 + L_2 + L_3$ is a direct sum

(B) $L_i \cap L_j = \{0\}$ for $i \neq j$, yet the sum is not direct

(C) $\dim(L_1 + L_2 + L_3) = 3$

(D) $L_1 \cup L_2$ is a subspace

10. Exam pattern NAT In $M_4(\mathbb{R})$, let S be the symmetric matrices and U the upper triangular matrices. Then $\dim(S + U) =$ _____.